

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihal Produk: **Di-n-butyltin dilaurate**  
 Product Description: **Di-n-butyltin dilaurate**  
 Cat No. : S71130  
 CAS No 77-58-7  
 Molecular Formula C32 H65 O4 Sn

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

**Company**

Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
 Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
 No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,  
 Selangor Darul Ehsan, Malaysia.  
 Main line: +60 3-5525 7888

**Supplier**

E-mail address Enquiry.my@thermofisher.com

**Emergency Telephone Number**

Tel: +03-5525 7888  
 CHEMTREC Malaysia **1-800-815-308** (Malay)  
 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                                      |                    |
|------------------------------------------------------|--------------------|
| Serious Eye Damage/Eye Irritation                    | Category 2 (H319)  |
| Skin Sensitization                                   | Category 1 (H317)  |
| Germ Cell Mutagenicity                               | Category 2 (H341)  |
| Reproductive Toxicity                                | Category 1B (H360) |
| Specific target organ toxicity - (single exposure)   | Category 1 (H370)  |
| Specific target organ toxicity - (repeated exposure) | Category 1 (H372)  |
| Acute aquatic toxicity                               | Category 1 (H400)  |
| Chronic aquatic toxicity                             | Category 1 (H410)  |

**Label Elements**


# SAFETY DATA SHEET

Di-n-butyltin dilaurate

Revision Date 01-Apr-2025

## Signal Word

## Danger

### Hazard Statements

H317 - May cause an allergic skin reaction  
H319 - Causes serious eye irritation  
H370 - Causes damage to organs  
H341 - Suspected of causing genetic defects  
H360 - May damage fertility or the unborn child  
H372 - Causes damage to organs through prolonged or repeated exposure  
H410 - Very toxic to aquatic life with long lasting effects

### Precautionary Statements

#### Prevention

P270 - Do not eat, drink or smoke when using this product  
P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P272 - Contaminated work clothing should not be allowed out of the workplace  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

#### Response

P333 + P313 - If skin irritation or rash occurs: Get medical advice/attention  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P337 + P313 - If eye irritation persists: Get medical advice/attention  
P362 + P364 - Take off contaminated clothing and wash it before reuse  
P308 + P311 - IF exposed or concerned: Call a POISON CENTER or doctor

#### Storage

P405 - Store locked up

#### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

### Other Hazards

Toxic to terrestrial vertebrates  
This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component            | CAS No  | Weight % |
|----------------------|---------|----------|
| Dibutyltin dilaurate | 77-58-7 | >95      |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical

# SAFETY DATA SHEET

Di-n-butyltin dilaurate

Revision Date 01-Apr-2025

|                                           |                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | attention is required.                                                                                                                                                                                                                                                                                                         |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| <b>Inhalation</b>                         | Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                               |

## **Most important symptoms and effects, both acute and delayed**

May cause allergic skin reaction. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

## **Indication of any immediate medical attention and special treatment needed**

**Notes to Physician** Treat symptomatically. Symptoms may be delayed.

## **SECTION 5: FIREFIGHTING MEASURES**

### **Extinguishing media**

#### **Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### **Special hazards arising from the substance or mixture**

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Do not allow run-off from fire-fighting to enter drains or water courses.

### **Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Metal oxides, Thermal decomposition can lead to release of irritating gases and vapors.

### **Advice for fire-fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## **SECTION 6: ACCIDENTAL RELEASE MEASURES**

### **Personal Precautions, Protective Equipment and Emergency Procedures**

Ensure adequate ventilation. Use personal protective equipment as required. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

### **Environmental precautions**

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent

# SAFETY DATA SHEET

Di-n-butyltin dilaurate

Revision Date 01-Apr-2025

product from entering drains. Local authorities should be advised if significant spillages cannot be contained. Should not be released into the environment.

## Methods and Material for Containment and Cleaning Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

## Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

### Conditions for Safe Storage, Including any Incompatibilities

Corrosives area. Store indoors. Keep containers tightly closed in a dry, cool and well-ventilated place.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component            | Malaysia | ACGIH TLV                                                         | OSHA PEL                                     |
|----------------------|----------|-------------------------------------------------------------------|----------------------------------------------|
| Dibutyltin dilaurate |          | TWA: 0.1 mg/m <sup>3</sup><br>STEL: 0.2 mg/m <sup>3</sup><br>Skin | (Vacated) TWA: 0.1 mg/m <sup>3</sup><br>Skin |

| Component            | European Union | The United Kingdom                                                            | Germany                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------|----------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dibutyltin dilaurate |                | STEL: 0.2 mg/m <sup>3</sup> 15 min<br>TWA: 0.1 mg/m <sup>3</sup> 8 hr<br>Skin | TWA: 0.0018 ppm (8 Stunden).<br>AGW - exposure factor 1<br>TWA: 0.009 mg/m <sup>3</sup> (8 Stunden).<br>AGW - exposure factor 1<br>TWA: 0.004 ppm (8 Stunden). MAK<br>can occur as vapor and aerosol at<br>the same time<br>TWA: 0.02 mg/m <sup>3</sup> (8 Stunden). MAK<br>can occur as vapor and aerosol at<br>the same time<br>Höhepunkt: 0.004 ppm<br>Höhepunkt: 0.02 mg/m <sup>3</sup> |

### Exposure Controls

#### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

#### Personal protective equipment

Eye Protection

Goggles

Hand Protection

Protective gloves

Skin and body protection

Long sleeved clothing

# SAFETY DATA SHEET

Di-n-butyltin dilaurate

Revision Date 01-Apr-2025

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

## Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

## Recommended Filter type:

Organic gases and vapours filter Type A Brown conforming to EN14387

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

## Environmental exposure controls

Prevent product from entering drains Do not allow material to contaminate ground water system Local authorities should be advised if significant spillages cannot be contained

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                |                       |
|----------------|-----------------------|
| Appearance     | Clear, Light yellow   |
| Physical State | Liquid Viscous liquid |
| Odor           | Characteristic        |
| Odor Threshold | No data available     |
| pH             | Not applicable        |

|                     |                             |                                   |
|---------------------|-----------------------------|-----------------------------------|
| Melting Point/Range | 16 - 18 °C / 60.8 - 64.4 °F |                                   |
| Softening Point     | No data available           |                                   |
| Boiling Point/Range | 205 °C / 401 °F             | @ 12 mmHg                         |
| Flash Point         | 149 °C / 300.2 °F           | Method - No information available |

|                          |                   |        |
|--------------------------|-------------------|--------|
| Evaporation Rate         | No data available |        |
| Flammability (solid,gas) | Not applicable    | Liquid |
| Explosion Limits         | No data available |        |

|                              |                          |             |
|------------------------------|--------------------------|-------------|
| Vapor Pressure               | No data available        |             |
| Vapor Density                | No data available        | (Air = 1.0) |
| Specific Gravity / Density   | 1.05                     |             |
| Bulk Density                 | Not applicable           | Liquid      |
| Water Solubility             | Insoluble                |             |
| Solubility in other solvents | No information available |             |

### Partition Coefficient (n-octanol/water)

|                      |         |
|----------------------|---------|
| Component            | log Pow |
| Dibutyltin dilaurate | 3.12    |

# SAFETY DATA SHEET

Di-n-butyltin dilaurate

Revision Date 01-Apr-2025

Autoignition Temperature No data available  
Decomposition Temperature No data available  
Viscosity No data available  
Explosive Properties No information available  
Oxidizing Properties No information available

Molecular Formula C32 H65 O4 Sn  
Molecular Weight 631.57

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Stable under recommended storage conditions.

### Possibility of Hazardous Reactions

Hazardous Polymerization Hazardous polymerization does not occur.  
Hazardous Reactions None under normal processing.

### Conditions to Avoid

Incompatible products. Excess heat.

### Incompatible Materials

Strong oxidizing agents.

### Hazardous Decomposition Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>). Metal oxides. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

##### (a) acute toxicity;

Oral Based on available data, the classification criteria are not met  
Dermal Based on available data, the classification criteria are not met  
Inhalation Based on available data, the classification criteria are not met

| Component            | LD50 Oral          | LD50 Dermal         | LC50 Inhalation |
|----------------------|--------------------|---------------------|-----------------|
| Dibutyltin dilaurate | 2071 mg/kg ( Rat ) | >2000 mg/kg ( Rat ) | -               |

(b) skin corrosion/irritation; Based on available data, the classification criteria are not met

# SAFETY DATA SHEET

Di-n-butyltin dilaurate

Revision Date 01-Apr-2025

(c) serious eye damage/irritation; Category 2

(d) respiratory or skin sensitization;

Respiratory  
Skin

Based on available data, the classification criteria are not met  
Category 1

May cause sensitization by skin contact

(e) germ cell mutagenicity;

Category 2

(f) carcinogenicity;

Based on available data, the classification criteria are not met  
There are no known carcinogenic chemicals in this product

(g) reproductive toxicity;  
Reproductive Effects  
Developmental Effects

Category 1B  
Experiments have shown reproductive toxicity effects on laboratory animals.  
Developmental effects have occurred in experimental animals.

(h) STOT-single exposure;

Category 1

Results / Target organs

Thymus.

(i) STOT-repeated exposure;

Category 1

Target Organs

Immune system, Thymus.

(j) aspiration hazard;

Based on available data, the classification criteria are not met

Symptoms / effects, both acute and delayed

Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

Endocrine Disrupting Properties

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity effects

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. Do not allow material to contaminate ground water system. The product contains following substances which are hazardous for the environment.

| Component            | Freshwater Fish                         | Water Flea            | Freshwater Algae | Microtox                 |
|----------------------|-----------------------------------------|-----------------------|------------------|--------------------------|
| Dibutyltin dilaurate | LC50 = 2 mg/L, 48h<br>(Oryzias latipes) | LC50 = 0.463 mg/L 48h |                  | EC50 = 0.576 mg/L 30 min |

Persistence and degradability

Persistence  
Degradation in sewage  
treatment plant

Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary  
May persist, based on information available.  
Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

# SAFETY DATA SHEET

Di-n-butyltin dilaurate

Revision Date 01-Apr-2025

## Bioaccumulative potential

May have some potential to bioaccumulate

| Component            | log Pow | Bioconcentration factor (BCF) |
|----------------------|---------|-------------------------------|
| Dibutyltin dilaurate | 3.12    | No data available             |

## Mobility in soil

Spillage unlikely to penetrate soil. The product is insoluble and sinks in water. The product evaporates slowly. Is not likely mobile in the environment due its low water solubility. Spillage unlikely to penetrate soil.

## Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

## Other adverse effects

No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

#### Waste from Residues/Unused Products

Should not be released into the environment Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

#### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point.

#### Other Information

Do not flush to sewer Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains Do not let this chemical enter the environment

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

UN-No UN3082  
Hazard Class 9  
Packing Group III  
Proper Shipping Name ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. Dibutyltin dilaurate

### Road and Rail Transport

UN-No UN3082  
Hazard Class 9  
Packing Group III  
Proper Shipping Name ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. Dibutyltin dilaurate

### IATA

UN-No UN3082  
Hazard Class 9  
Packing Group III  
Proper Shipping Name ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. Dibutyltin dilaurate

## Special Precautions for User

No special precautions required

## SECTION 15: REGULATORY INFORMATION



# SAFETY DATA SHEET

Di-n-butyltin dilaurate

Revision Date 01-Apr-2025

## Safety, health and environmental regulations/legislation specific for the substance or mixture

### International Inventories

China X = listed Australia U.S.A. (TSCA) Canada (DSL/NDSL) Europe (EINECS/ELINCS/NLP) Australia (AICS) Korea (KECL) China (IECSC) Japan (ENCS) Philippines (PICCS) Taiwan (TCSI) Japan (ISHL) New Zealand (NZIoC) Japan (ISHL)

| Component            | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|----------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Dibutyltin dilaurate | 201-039-8 | X    | X   | X     | X    | X    | X     | X    | KE-09969 |

### National Regulations

#### Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

#### Prepared By

Health, Safety and Environmental Department

#### Revision Date

01-Apr-2025

#### Revision Summary

Initial Release.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

### Disclaimer

# SAFETY DATA SHEET

Di-n-butyltin dilaurate

Revision Date 01-Apr-2025

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The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**